



Using Conditional Diffusion Models for Data-Driven Force Identification of Structural Dynamic Systems Under Uncertainty

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Motivation

Force identification from sparse sensing

- Structural health monitoring relies on sensor networks, yet the excitation may be unknown or unmeasured in practice.
- Recovering it from sparse, potentially corrupted measurements is challenging, as the solution is non-unique and the observations are incomplete.

Gaps in current inverse methods

- Classical inverses are limited to fixed observation matrices and specified noise models.
- Deep regressors trained with MSE converge to the conditional mean, which can smooth high-frequency components.
- In SHM, diffusion models have primarily been used for forward modeling, not to recover the excitation.

Objective

InvSHMDiff learns the conditional distribution of ground excitation given incomplete accelerometer responses; the same ensemble also yields relative uncertainty and virtual-sensing estimates.

Methodology: InvSHMDiff^[1]

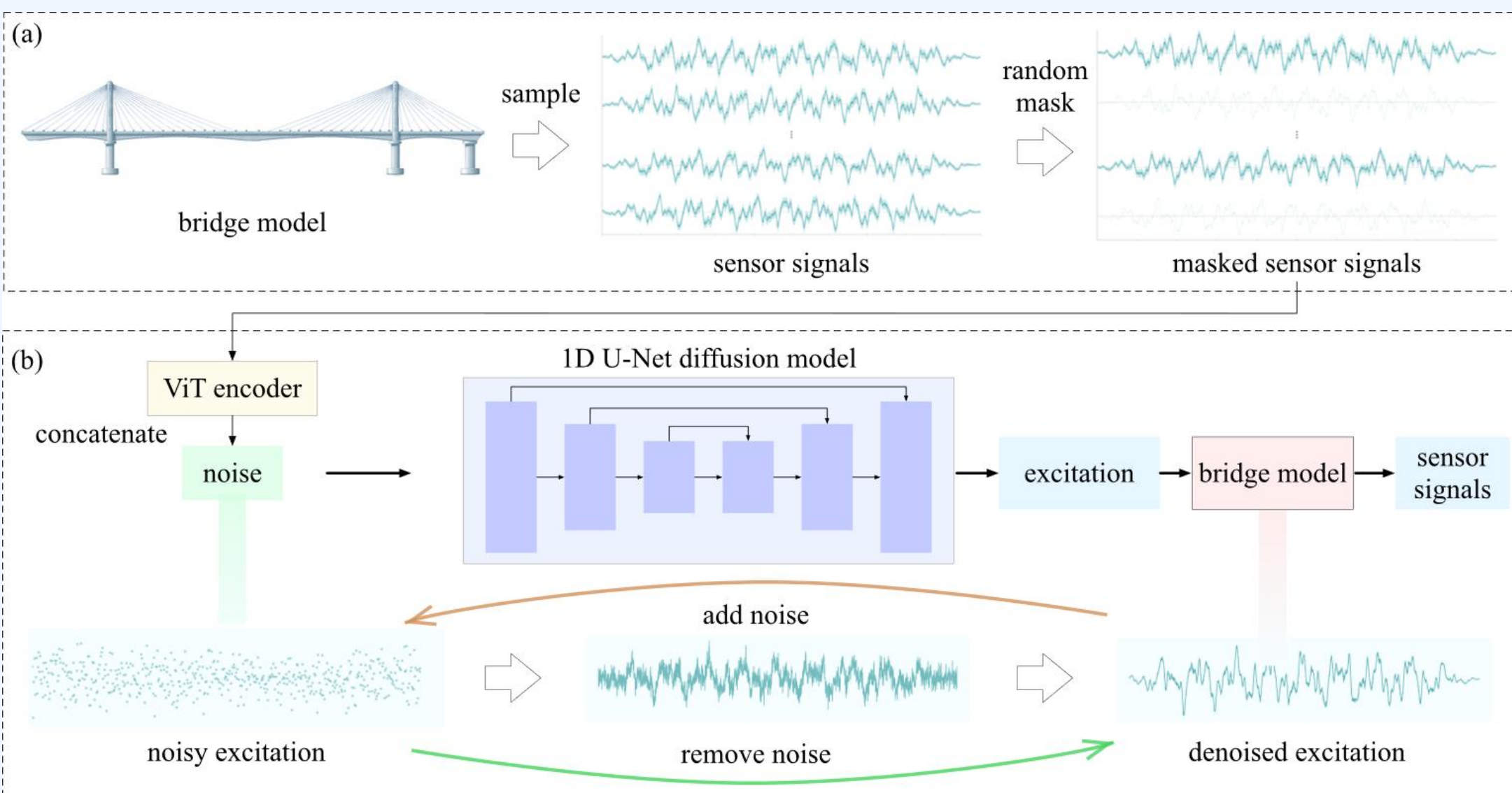


Fig. 1. Overview of InvSHMDiff: (a) training pairs with random sensor masks and (b) conditional denoising of the excitation.

$$\mathbf{y} = \mathcal{A}(\mathbf{x}) + \boldsymbol{\eta} \implies \text{learn } p_{\theta}(\mathbf{x}_0 | \mathbf{y}) \quad \mathcal{L}(\theta, \phi) = \mathbb{E} \left[\lambda(t) \| f_{\theta}([\mathbf{x}_t; \mathcal{E}_{\phi}(\mathbf{y}, \mathbf{M})], t) - \mathbf{v}_t \|_2^2 \right], \quad \mathbf{v}_t = \sqrt{\bar{\alpha}_t} \boldsymbol{\epsilon} - \sqrt{1 - \bar{\alpha}_t} \mathbf{x}_0$$

(1) Sensor-level masking

Entire channels removed at random each step, with a binary mask channel

(2) ViT sensor encoder

Self-attention across 832 sensor-time tokens from the 26 channels

(3) Conditional denoising

1D U-Net with v-prediction; mean and spread of 30 DDIM samples

The FE model is used only to generate the training pairs and is never evaluated during inversion.

Numerical Testbed

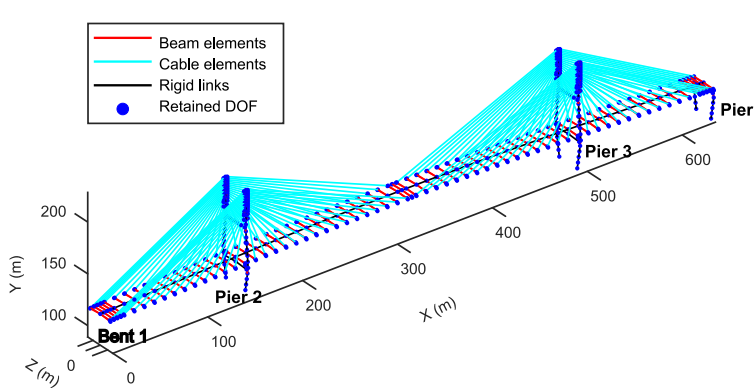


Fig. 2. ASCE benchmark FE model (579 nodes) of the cable-stayed Bill Emerson Memorial Bridge.

Simulation archive

- 12,000 Kanai-Tajimi ground motions
- 26 vertical deck accelerometers, 10 s records
- 10,000 training and 2,000 test records
- Compared with one model-driven baseline (mask-aware, FE-based Tikhonov) and four learned baselines (BiGRU, BiLSTM, cGAN, and TCN) sharing the same data and masking protocol.

Robustness to Sensor Loss

Mean R^2 remains above 99.79% even when 13 of the 26 channels are missing. Both the amplitude and phase of the ground excitation are recovered at every mask ratio.

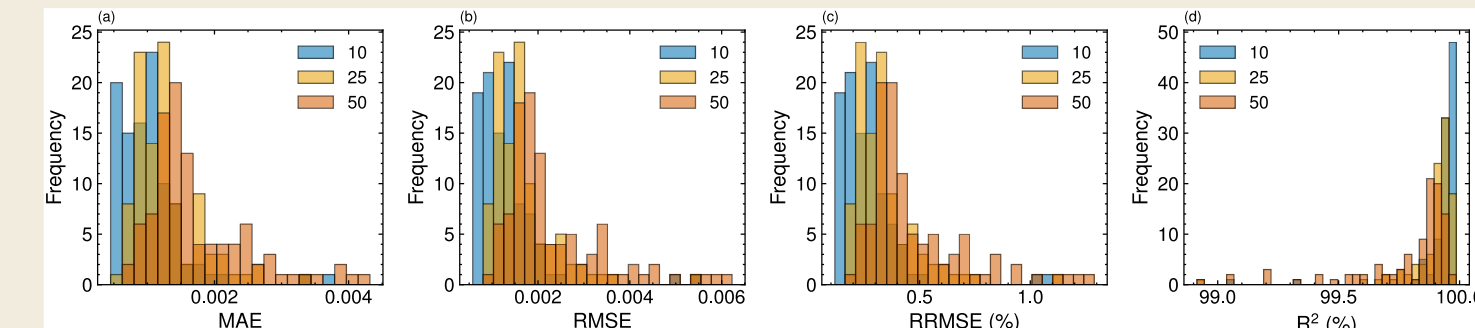


Fig. 3. Per-record error distributions at matched mask ratios of 10%, 25%, and 50% (one model per ratio).

Robustness to Measurement Noise

InvSHMDiff is the most accurate method under colored noise at 30 dB and below, and the strongest learned method under white noise.

At 20 dB, it reaches **99.35%** under colored noise and **95.08%** under white noise, second only to the FE-based Tikhonov inverse (**98.16%**).

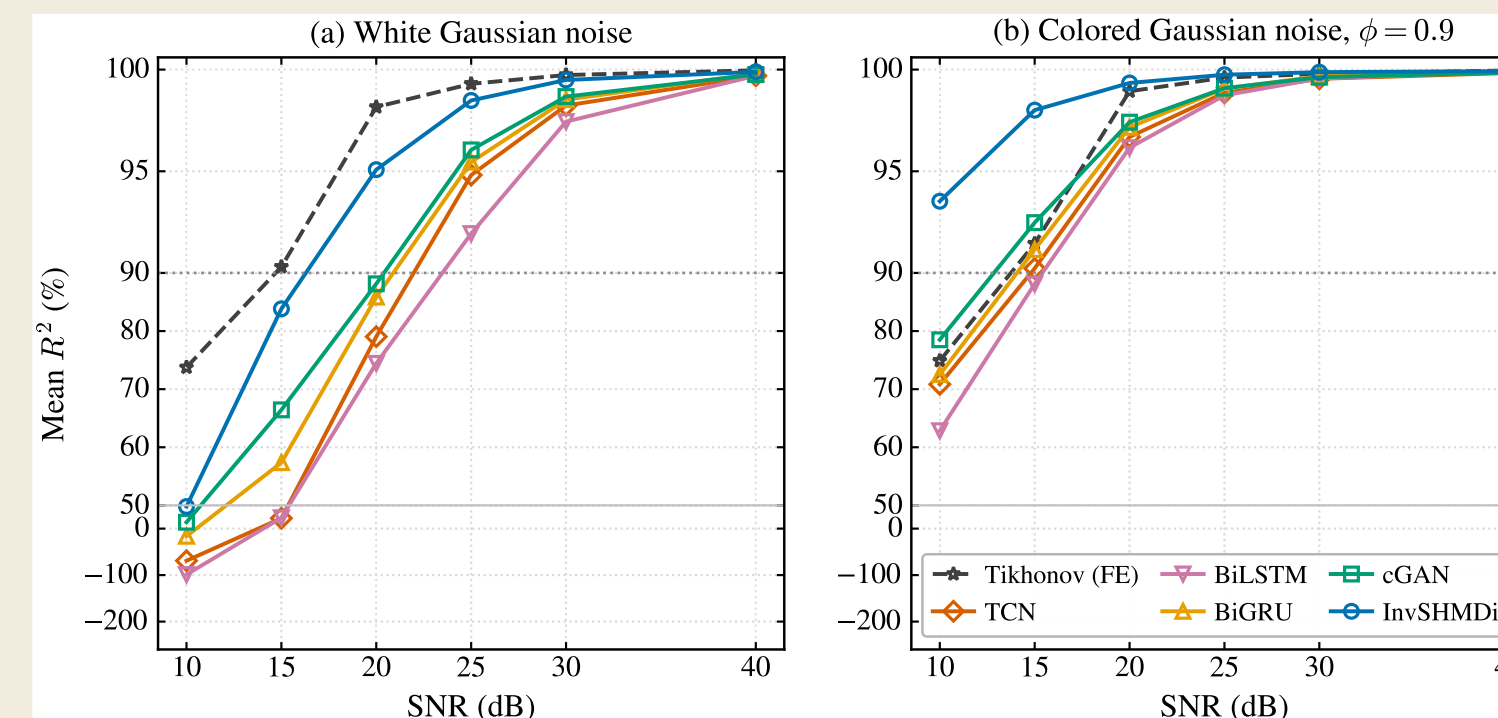


Fig. 4. Mean R^2 over 100 held-out records against SNR ($r = 0.10$; all models trained on clean data; Tikhonov strength per record by the L-curve); vertical axis expanded above 90% and compressed below 50%.

Recorded Ground Motions, Simulated Bridge Responses

InvSHMDiff leads on difficult records (5th-percentile R^2 of 88.44% against 82.92% for the TCN) and gives the lowest PGA, Arias-intensity, and response-spectrum errors (5.51%, 10.70%, and 8.50%).

Its mean R^2 of 95.59% is effectively tied with the TCN (95.66%).

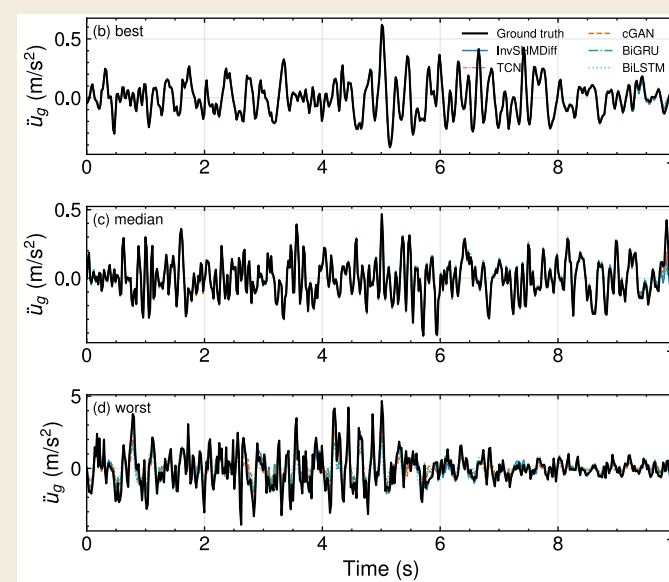


Fig. 5. Recorded and reconstructed motions for the best, median, and worst records.

76 recorded motions

- Eight earthquakes recorded in California (M_w 5.1 to 7.2) plus El Centro, Kobe, and Morgan Hill
- Up to 0.76 g, 6.4 times the largest training PGA
- Recorded earthquake inputs; bridge responses simulated with the FE model
- From the lowest to the highest PGA tercile, R^2 drops 4.3% for InvSHMDiff against 7.2 to 21.3% for the learned baselines.

Relative Uncertainty and Model-Assisted Virtual Sensing

The ensemble spread correlates with per-record RMSE (Pearson 0.871 to 0.959), supporting its use as a relative indicator of reconstruction quality.

The spread also ranks records by quality (Spearman 0.669 to 0.878), indicating which reconstructions to distrust without ground truth.

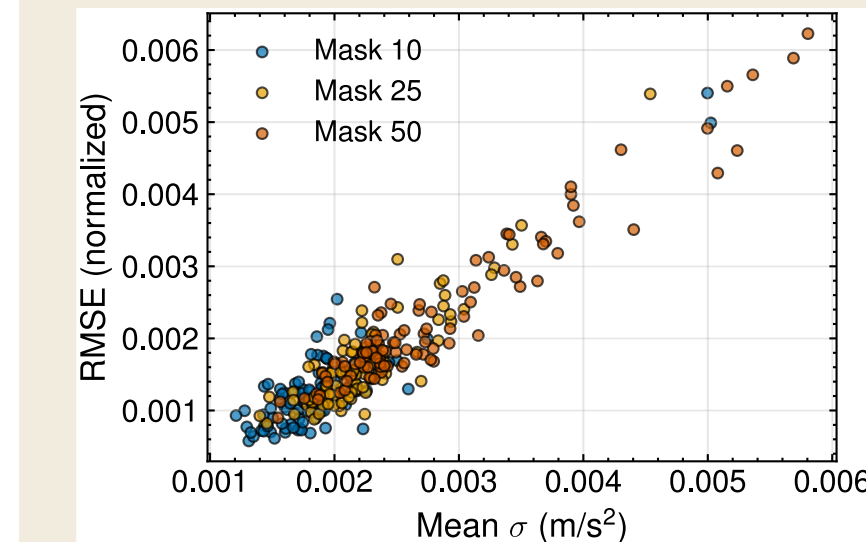


Fig. 6. Per-record ensemble spread against RMSE.

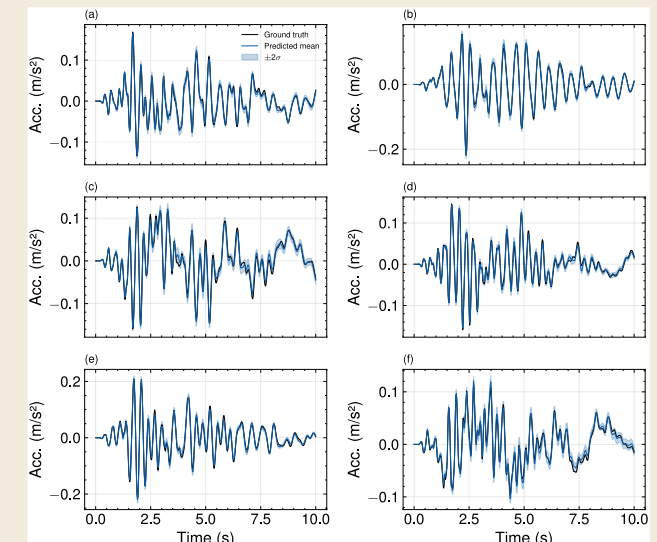


Fig. 7. Virtual sensors with $\pm 2\sigma$ bands.



- Mean R^2 is 98.95% at virtual channels against 99.03% at observed channels, even with the excitation inferred from half the channels.
- Under FE model mismatch, InvSHMDiff degrades far more slowly than Tikhonov: at 10% stiffness error it retains an R^2 of 59.8 to 64.3% against 29.5 to 31.3%.

Transfer to Nonlinear Dynamics



At 20 dB noise, fine-tuned InvSHMDiff is the most accurate (R^2 of 86.90% against 81.30% for the TCN), while linear Tikhonov fails (0.18%).

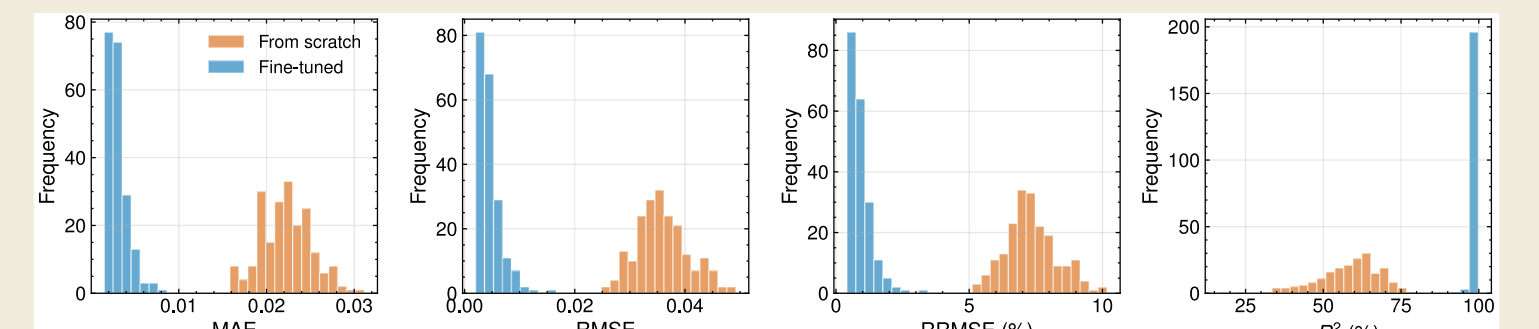


Fig. 8. Per-record errors on the 200 nonlinear test records, fine-tuned against trained from scratch.

- On clean data, fine-tuning on 800 records reaches an R^2 of 99.25% against 58.15% from scratch; baselines fine-tuned the same way come within 0.7%.

Conclusions

On the benchmark cable-stayed bridge, InvSHMDiff:

- recovers the amplitude and phase of the excitation with half of the sensors unavailable;
- is the most accurate under colored noise, second only to FE-based Tikhonov under white noise, and leads on difficult recorded motions and all three physical measures;
- yields relative uncertainty indicators and virtual-sensing estimates from one ensemble;
- adapts to nonlinear dynamics by fine-tuning and is the most accurate under noise.

Future work will pursue experimental validation on physical structures, cross-structure transfer, and recalibration of the uncertainty indicators.

Reference

J. Tang, D. An, P. T. Brewick, C. Wang, Using conditional diffusion models for data-driven force identification of structural dynamic systems under uncertainty, *Advanced Engineering Informatics* (2026), accepted (Q1 TOP, IF 11.5).

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